

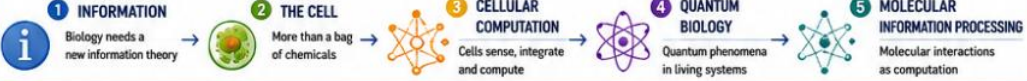


RETHINKING BIOLOGY: ARTICLE 5 FROM CHEMISTRY TO INFORMATION PROCESSING

COULD MOLECULAR INTERACTIONS THEMSELVES BE A FORM OF COMPUTATION?

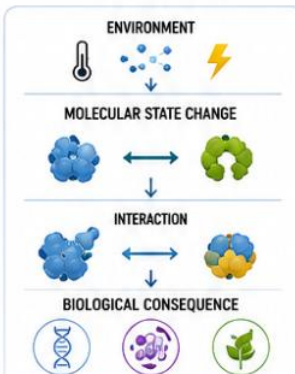
Molecules don't just react. They sense, change state, interact and respond.
What if life's computation begins at the molecular level?

THE JOURNEY SO FAR



MOLECULES AS INFORMATION PROCESSORS

- Molecules occupy physical states.
- Interactions change those states.
- States influence future interactions.
- That is information processing.



Information processing does not require an electronic computer. It can be embodied in physical state changes.

FROM REACTIONS TO COMPUTATION

Molecular pathways form networks with:



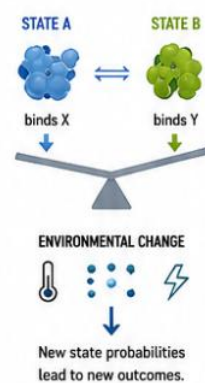
Organization of interactions creates computational properties.

The computation may not reside in the molecule. It may reside in the interaction.

THOUGHT EXPERIMENTS

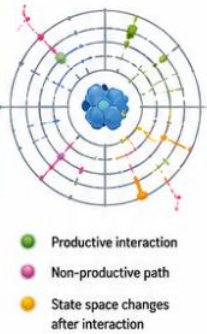
1. THE TWO-STATE PROTEIN

A protein exists in two states: A and B. Environment shifts the balance. Different interactions follow.



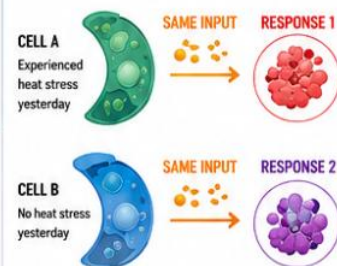
2. THE MOLECULAR MAZE

Proteins explore a vast landscape of configurations and interactions. Useful paths emerge from physical dynamics.



3. SAME INPUT, DIFFERENT HISTORY

Identical cells. Same input. Different history → different response. Computation depends on state and history.



Biological computation is:
Context-dependent • History-dependent
Adaptive • Substrate changes while computing

COULD QUANTUM PHYSICS CONTRIBUTE?

Molecules are quantum systems. But does quantum behavior provide a functional advantage in biological information processing?



THE MOLECULAR QUANTUM-SEARCH HYPOTHESIS

Interacting molecules may explore many possible quantum states. Their interaction could help search for biologically relevant configurations more efficiently.



CAUTION

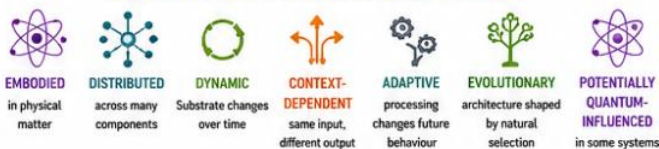
- Superposition = computation
- Quantum nature = quantum function
- Hypothesis, not established fact

WHAT WOULD COUNT AS EVIDENCE?

- ✓ Quantum behavior provides a measurable functional advantage.
- 🔍 Behavior cannot be explained by classical models but fits quantum information models.
- 📈 Scaling or efficiency patterns consistent with quantum information processing.
- 🧪 New predictions that distinguish quantum vs classical explanations.

Science progresses by making predictions that can be proven wrong. That is how hypotheses become theories.

WHAT MAKES BIOLOGICAL COMPUTATION UNIQUE?



A POSSIBLE RESEARCH PROGRAMME

- 1 Identify biological molecular systems where information processing is tied to function.
- 2 Characterize their states and transitions.
- 3 Model with classical and (where appropriate) quantum models.
- 4 Design experiments that distinguish the competing models.
- 5 Let data decide.



THE BIGGER PICTURE: FROM CHEMISTRY TO ADAPTIVE INFORMATION PROCESSING



KEY TAKEAWAY

Molecular interactions might be more than chemistry. They may carry, transform and process information. That possibility opens a new way to understand life.

WHERE WE ARE IN THE SERIES



We have now travelled quite a long way.

We began this series by asking whether biology needs to think differently about information.

We then looked inside the cell and questioned whether it is really adequate to describe it simply as a collection of chemicals.

We explored the possibility that cells can sense, integrate information, respond to signals and change their internal states—in other words, that cellular processes can be viewed through the lens of computation.

Then we entered the world of quantum biology and saw that some biological processes may involve genuinely quantum phenomena.

Each step changed the question slightly.

From:

What is life made of?

to:

How is life organized?

to:

How does life process information?

to:

What role might quantum physics play in that process?

Now we can go one step deeper.

What happens if we stop looking at the cell as our basic computational unit?

What if we look at the **molecules themselves**?

Could molecular interactions be more than chemical reactions?

Could they also be **information-processing events**?

And, more provocatively:

Could some of the computational organization of life emerge directly from the physical interactions among molecules?

This is where the discussion becomes more personal.

Some of the ideas I will explore in this article arise from hypotheses I proposed in my earlier work. I want to present them differently here—not as established facts, but as questions that may be worth investigating.

That distinction is important.

The purpose of this article is not to claim that we have discovered how biological computation works. It is to ask whether there may be a level between **chemistry and computation** that we have not yet fully understood.

The molecule as more than a chemical object

When we learn molecular biology, we naturally classify molecules according to their functions.

A protein may be an enzyme.

Another may be a receptor.

Another may be a transcription factor.

A membrane protein may function as a channel.

DNA stores genetic information.

RNA can carry and regulate information.

These descriptions are useful because they tell us what molecules do.

But consider a protein more carefully.

A protein is not a rigid object.

It is a dynamic physical system.

Its atoms are continuously moving.

Its structure fluctuates.

Different parts of the molecule can move relative to one another.

It can occupy different conformational states.

Some of those states may bind one molecule more effectively than another.

Some may expose a binding site.

Others may hide it.

A chemical modification can change its behaviour.

A change in its environment can alter its state.

A binding event can change its conformation.

That conformational change can influence another molecule.

And so the process continues.

We can describe this entirely in chemical language.

But we can also describe it in terms of **states and transitions**.

The protein begins in one physical state.

Something happens.

It moves to another state.

That state determines what happens next.

This is already beginning to sound like information processing.

A very simple thought experiment

Imagine a protein that can occupy two distinguishable states.

Call them **A** and **B**.

In state A, the protein interacts strongly with molecule X.

In state B, it interacts strongly with molecule Y.

Now suppose the surrounding environment changes.

Perhaps an ion concentration changes.

Perhaps another protein binds.

Perhaps the temperature changes.

Perhaps a signalling molecule modifies the protein.

The protein responds by changing the relative probability of occupying A or B.

Nothing resembling a conventional computer has appeared.

There is no processor.

There is no software.

There is no binary code.

But the protein has undergone a transformation:

environment → molecular state → interaction → biological consequence

The physical state of the protein carries information about its environment.

Its state influences what it does next.

In that broad sense, the protein participates in information processing.

The point is not that a protein is a miniature laptop.

It is not.

The point is more fundamental:

Information processing does not necessarily require an electronic computer. It can be embodied in physical state changes.

Earlier work in this area considered precisely this connection between physical molecular states and computation. A protein occupying distinguishable states can be regarded, in a simplified information-

theoretic representation, as registering information; when it changes state, that information is transformed.

This leads to a question that I find increasingly interesting:

If molecules can carry and transform information through their physical states, could molecular interactions themselves constitute a form of computation?

The chemistry does not disappear

There is an important qualification.

Saying that molecular interactions process information does **not** mean that chemistry has somehow become irrelevant.

Quite the opposite.

The chemistry is the mechanism.

A protein binds another molecule because of physical and chemical interactions.

A receptor changes conformation because of molecular forces.

An enzyme changes the rate of a reaction because of its molecular structure.

A signalling cascade operates through biochemical interactions.

Nothing about the information-processing perspective invalidates these mechanisms.

Instead, it gives us another level at which to describe them.

Consider a receptor on a cell membrane.

At the molecular level:

ligand binds receptor.

At the cellular level:

cell receives a signal.

At the information level:

an external state is transformed into an internal representation.

At the computational level:

the system has transformed an input into a new internal state.

All four descriptions can refer to the same physical event.

The question is not which description is "correct."

The question is:

Which level of description helps us understand the phenomenon we are trying to explain?

From reaction to information

Traditional molecular biology often describes pathways as chains of reactions.

A molecule interacts with another molecule.

A product is formed.

That product interacts with another molecule.

Eventually, a cellular response occurs.

This description is powerful.

But when the network becomes sufficiently complex, something interesting happens.

The pathway begins to look less like a sequence and more like a **computational network**.

Several inputs converge.

Some signals reinforce one another.

Others cancel each other.

Feedback modifies the response.

Thresholds determine whether a pathway switches on.

Different combinations of inputs can produce different outputs.

The network can retain a memory of previous states.

Now consider what has happened.

The molecules have not stopped being molecules.

The reactions have not stopped being reactions.

But the **organization of the reactions** has acquired computational properties.

This is one of the central ideas of systems biology.

The properties of a biological system cannot always be inferred simply by looking at its components one at a time.

The relationships among the components matter.

And those relationships carry information.

The molecular search problem

Now let us consider something even more basic.

A protein inside a cell may need to interact with another molecule.

But the cell is extraordinarily crowded.

There may be thousands of different molecular species.

There are many potential interactions.

Only a fraction of them are biologically useful.

The protein therefore has, metaphorically speaking, a **search problem**.

It must find an appropriate partner.

The conventional molecular explanation is already sophisticated.

Molecules diffuse.

They collide.

Their structures and chemical properties determine whether productive interactions occur.

Binding affinity matters.

Electrostatics matter.

Hydrophobic interactions matter.

Molecular shape matters.

The surrounding environment matters.

There is no need to replace this framework with something more exotic simply because the word "search" sounds computational.

But the computational analogy raises an interesting question.

Suppose we think not only about where a molecule is in physical space, but about the enormous number of **possible molecular configurations and interactions** available to it.

The molecule is effectively moving through a landscape of possibilities.

Some configurations are favourable.

Some are unfavourable.

Some lead to useful interactions.

Most do not.

Could this process be understood as a form of physical search?

And if so:

What determines the efficiency of that search?

That is a much more interesting question than simply saying that molecules "find" one another.

Thought experiment: the molecular maze

Imagine a protein entering a vast molecular maze.

At every moment it has many possible configurations.

Some expose a binding surface.

Others hide it.

Some allow interaction with one molecular partner.

Others allow interaction with another.

Most configurations lead nowhere useful.

The protein is not consciously searching.

It does not have a map.

There is no little computer inside it deciding which direction to take.

Yet its physical dynamics determine the paths it can explore.

Now imagine that the protein encounters another molecule.

The interaction changes its conformational landscape.

Some possible states become more likely.

Others become inaccessible.

The molecule has effectively changed the space through which it can search.

In this picture, an interaction is not simply:

molecule A + molecule B → binding

It is also:

interaction → change in state space → change in future possibilities.

That is an information-processing perspective.

The interaction changes what the system can do next.

When search becomes computation

Search is one of the most familiar problems in computer science.

Given a large space of possibilities, identify the relevant one.

A computer performs a search according to an algorithm.

A biological molecule does not necessarily have an algorithm in that conventional sense.

But the physical system can nevertheless explore a space of possibilities.

This raises an important conceptual distinction.

Algorithmic computation is one kind of computation.

Physical computation is broader.

A physical system can transform information through its dynamics without executing a symbolic program.

A biological system may therefore compute without ever representing information in the form of numbers or symbols.

This is one reason the language of computation can be useful in biology.

It allows us to ask questions about:

search,

optimization,

state transitions,

memory,

feedback,

recognition,

decision,

and adaptation.

But it also creates a danger.

If we call every physical process "computation," the concept becomes meaningless.

The challenge is therefore to identify **specific computational properties** that are biologically relevant and measurable.

What does a molecule actually "know"?

There is another way of asking the same question.

Suppose a protein binds only to a particular class of molecules.

We might say that the protein "recognizes" them.

But what does recognition mean physically?

The protein does not possess a mental representation of the molecule.

It does not know its name.

It does not consciously identify it.

Recognition occurs because the physical properties of the two molecules permit a particular interaction.

Yet from an information perspective, something interesting has happened.

The protein has distinguished one class of molecular states from another.

It has responded differently to different inputs.

In other words, **physical discrimination has occurred**.

This may be one of the most primitive forms of biological information processing.

Before there were nervous systems, there were molecules capable of distinguishing one molecular environment from another.

Before there were brains, there were molecular systems capable of selective interaction.

Before there were computers, there were physical systems transforming states.

Perhaps computation in biology therefore has a much older history than we usually imagine.

Could proteins have computational properties?

This question has been part of my own work for some time.

In one earlier study, I explored the idea of assigning a theoretical computational capacity to proteins based on physical limits on information processing.

The analysis considered proteins involved in heat, drought and salinity stress and compared their estimated maximum computational capacities. The study reported statistically significant differences between some groups of stress-associated and non-associated proteins.

The calculation was based on physical limits relating energy to the maximum rate at which a physical system can perform state transitions or logical operations. The work explicitly treated these as **maximum computational capacities**, not measurements of the actual computation being performed by the proteins.

That distinction is important.

A physical upper bound is not the same thing as observed computational activity.

A protein may have the physical capacity for an enormous number of state transitions without actually using that capacity.

The work therefore should be regarded as an attempt to ask whether concepts from information processing and computational physics can be applied quantitatively to biological molecules.

The more interesting question that follows is:

Can such measures eventually be connected to measurable biological function?

That remains open.

From individual molecules to molecular networks

A single molecule is only the beginning.

Suppose thousands of molecules interact.

Now the number of possible system states becomes enormous.

Some interactions are mutually reinforcing.

Others are inhibitory.

Some interactions create feedback loops.

Some generate oscillations.

Some produce stable states.

Some make the system sensitive to small changes.

Others make it robust against noise.

The resulting network can behave in ways that are difficult to predict simply by studying individual molecules.

This is where a crucial idea emerges:

The computation may not reside in the molecule. It may reside in the interaction.

A molecule by itself has a set of possible states.

An interaction changes those possibilities.

A network of interactions creates a landscape of possible system states.

The network then becomes a physical system capable of transforming information.

This is very different from the architecture of a conventional computer.

In a conventional computer, the architecture is designed first.

In biology, the architecture emerges through evolution.

Evolution as a designer without a designer

This may help explain why biological systems can be so computationally sophisticated.

Evolution does not begin with a specification saying:

"Build a system capable of integrating five environmental signals."

Instead, variation produces many molecular configurations.

Selection preserves configurations that improve survival and reproduction.

Over enormous periods of time, networks become more elaborate.

Feedback mechanisms are added.

Interactions become specialized.

Redundant pathways provide robustness.

Alternative pathways provide flexibility.

The result can look as though it were engineered.

But there is no central engineer.

Evolution is effectively searching through a vast space of biological possibilities.

This gives us an interesting parallel.

At the molecular level, a biological system may perform physical search.

At the evolutionary level, populations perform an entirely different kind of search.

Both involve enormous spaces of possibilities.

Both involve constraints.

Both involve selection.

Both can generate solutions without an explicit designer.

Perhaps computation is therefore woven into biology at several different levels.

A second thought experiment: the same input, different history

Imagine two genetically identical cells.

Give them exactly the same external signal.

In a simple input-output machine, we might expect the same output.

But now imagine that Cell A experienced heat stress yesterday, while Cell B did not.

Their current molecular states may be different.

Some proteins may have been modified.

Some genes may have different expression levels.

Metabolism may have changed.

Signalling pathways may have different sensitivities.

The same external input can therefore produce a different response.

What does this tell us?

The biological system is not computing:

input → output

It is more like:

input + internal state + history → new state.

The computation depends upon the system's history.

This is one reason biological computation is so different from the simple computation performed by a calculator.

The machine itself changes.

The computation changes the machine that will perform the next computation.

The cell as a changing computational substrate

A biological system is therefore not merely a computer that processes information.

It is a computer that **changes while it computes**.

The processing modifies the physical substrate.

The modified substrate changes future processing.

This produces a feedback loop:

information → state change → altered processing → new information.

That is adaptation.

And adaptation may be one of the defining characteristics of biological information processing.

This idea also provides a bridge to artificial intelligence.

Modern AI systems can modify internal parameters during learning.

Biological systems modify their molecular states, networks and structures through development, experience and environmental interaction.

The comparison is not exact.

But it raises an important question:

Could biological computation be fundamentally adaptive rather than merely algorithmic?

We will return to this question in Article 6.

For now, there is an even deeper issue.

What does quantum biology add?

In the previous article, we examined examples in which quantum phenomena may have functional relevance to biological processes.

That raises a new possibility.

If biological molecules are quantum physical systems—and they necessarily are at the fundamental level—could their quantum properties contribute to information processing?

This is a much stronger question than:

"Is biology quantum?"

Everything molecular is quantum mechanical at the fundamental level.

The interesting question is:

Does the specifically quantum character of a biological system contribute functionally to the way information is processed?

That is much harder to establish.

And it is here that my earlier work proposed a more speculative possibility.

The molecular quantum-search hypothesis

In earlier work on quantum computation and molecular interactions, I proposed that interacting molecules in a living cell could be conceptualized as systems containing multiple possible quantum states and that their interaction might function as a search for biologically relevant configurations.

In that formulation, biomolecules were described as "databases" of quantum superpositions that could interact in a way analogous to quantum search.

This is an intriguing idea.

But it is also a strong claim.

So I would like to reformulate it more cautiously here.

Instead of saying:

"Biological molecules are quantum databases."

we can ask:

"Could the quantum state space of a biomolecule contribute to the search for biologically relevant interactions?"

That change is not merely rhetorical.

It turns a declaration into a research question.

Thought experiment: a quantum molecular search

Imagine again the protein in its molecular maze.

In a classical description, the protein occupies a particular physical state at any moment, and its dynamics determine which states it can reach.

Now imagine that, under appropriate physical conditions, several possible molecular states contribute coherently to the system's quantum description.

Could the molecule's interaction with another molecule alter this state space in a way that helps identify a biologically relevant configuration?

We should be extremely careful here.

This thought experiment does **not** show that such a mechanism exists.

It does not demonstrate that proteins maintain useful quantum coherence in the crowded cellular environment.

It does not demonstrate that cells execute Grover's algorithm or any other known quantum algorithm.

It does not demonstrate quantum advantage in molecular recognition.

It simply asks whether there could be a connection between:

quantum state space

and

biological search.

If such a connection exists, we should eventually be able to identify physical signatures of it.

If it does not, experiments should be able to rule it out.

That is exactly how a speculative idea should be treated.

Superposition is not automatically computation

This distinction deserves emphasis.

A quantum system can exist in a superposition.

That does not automatically make it a quantum computer.

A quantum system can be entangled.

That does not automatically mean it is performing a useful computation.

A molecule can be quantum mechanical.

That does not automatically mean that its biological function depends upon quantum information processing.

A quantum computer requires more than quantum states.

It requires controlled operations, information encoding, evolution of the computational state and meaningful readout.

Therefore, if we want to claim that biology performs quantum computation, we need to identify the biological equivalents of these elements.

What is the information being encoded?

What constitutes an operation?

What controls the evolution?

What is the output?

What constitutes an advantage over a classical mechanism?

These are difficult questions.

But they are the right questions.

What would count as evidence?

Suppose the molecular quantum-search hypothesis were true.

What would we expect to observe?

One possibility would be a biological process in which quantum coherence or another distinctly quantum phenomenon produces a measurable functional advantage.

Another would be a molecular system whose behaviour cannot be adequately explained by a classical model but is naturally explained by a quantum-information model.

Another possibility would be the discovery of scaling behaviour: as the number of relevant molecular states increases, the biological process could display an efficiency pattern characteristic of quantum information processing.

These are not predictions that have been established by the work discussed here.

They are examples of the kind of predictions that would need to be developed.

This is where the hypothesis becomes a potential research programme.

The classical challenge

But there is an equally important question.

What if a classical model explains everything?

Imagine that we take a molecular recognition process.

We build a detailed classical model incorporating diffusion, molecular structure, electrostatics, conformational dynamics, binding kinetics and environmental noise.

The model reproduces the experimental observations.

It predicts the behaviour of the biological system.

And it does so without requiring quantum computation.

Would we still need the quantum hypothesis?

Probably not.

That would not be disappointing.

It would be a successful scientific explanation.

A quantum hypothesis should not be accepted merely because quantum mechanics is fundamental.

It should earn its place by explaining something that the classical model cannot explain adequately, or by producing new predictions that can be tested.

This is particularly important for the direction I am proposing.

The purpose of a new biological theory is not to make biology more mysterious. It is to make biological phenomena more understandable.

A third thought experiment: two explanations

Imagine that we observe a biological process with extraordinary efficiency.

Model A explains it entirely using conventional molecular interactions.

Model B explains it using a quantum-information mechanism.

Both models reproduce the existing data.

Which one should we choose?

At this point, we should choose neither.

We need another experiment.

If Model B predicts something Model A does not, perform the experiment.

If Model A survives and Model B fails, discard Model B.

If Model B succeeds where Model A fails, then the quantum hypothesis becomes much more interesting.

This is the difference between **an appealing story** and **a scientific theory**.

A theory earns credibility through discrimination between possibilities.

The biological meaning of information

There is another reason this molecular perspective matters.

In conventional information theory, information is often associated with symbols, messages or states.

Biology uses a much richer vocabulary.

Information may be encoded in:

DNA sequence,

RNA abundance,

protein concentration,

protein conformation,

post-translational modification,

molecular location,

membrane potential,

metabolic state,

timing,

spatial organization,

and interactions.

A molecule therefore does not simply "contain information."

Its **state participates in a network of relationships**.

The information may lie partly in the molecule and partly in the relationship between the molecule and its environment.

This is very different from the idea of information as something stored in a passive container.

Biological information is dynamic.

It is continually being created, transformed, transmitted and destroyed.

Information is not the same as data

This distinction is useful.

A DNA sequence can be regarded as data.

But its biological meaning depends upon context.

The same sequence can have different consequences depending upon:

where it is expressed,

when it is expressed,

which proteins are present,

what regulatory elements are active,

what the cell's state is,

and what environmental conditions exist.

Thus biological information is not simply stored data.

It is **data interpreted by a dynamic system**.

And interpretation itself can be viewed as a physical process.

A transcription factor recognizes a sequence.

A receptor responds to a ligand.

An enzyme responds to its substrate.

A signalling network integrates multiple inputs.

At each stage, physical interactions transform potential information into functional consequences.

This may be one of the deepest reasons why information theory and biology increasingly need one another.

From molecules to networks

Let us now move upward again.

A molecule changes state.

That change affects another molecule.

Several molecules form a network.

The network has feedback.

Feedback creates stable and unstable states.

The system responds to inputs.

Its response depends on history.

The network changes its own future behaviour.

We have now moved from:

molecular chemistry

to:

molecular information processing

to:

network computation

to:

cellular adaptation.

The higher-level cellular computation discussed earlier in this series may therefore have a deeper physical foundation.

The cell does not suddenly become a computer at the cellular level.

The computational organization may emerge progressively from molecular interactions.

This is the possibility I find most intriguing.

Perhaps computation is emergent

Consider a single neuron.

It is complex.

But the behaviour of the brain cannot be explained simply by describing one neuron.

Now consider a single cell.

It is extraordinarily complex.

But its behaviour cannot be explained simply by describing one protein.

The same principle may apply at every level.

Individual components interact.

Interactions create networks.

Networks create system states.

System states create new capabilities.

Computation may therefore be an **emergent property of organization**.

This would explain why it is misleading to search for a tiny "computer" inside the cell.

There may be no single computational object.

There may instead be a hierarchy of physical processes whose collective organization produces computation.

What makes biological computation different?

If this perspective is correct, biological computation differs from conventional computation in several important ways.

It is **embodied** because the computation is implemented directly in physical matter.

It is **distributed** because many molecular components contribute simultaneously.

It is **dynamic** because the computational substrate changes over time.

It is **context-dependent** because the same input can produce different outputs depending on internal state.

It is **adaptive** because processing changes the future state of the system.

It is **evolutionary** because the architecture itself has been shaped by natural selection.

And perhaps, in some systems, it may also be **quantum-influenced**.

That final possibility remains the most speculative.

But it is now placed within a much broader framework.

Why this matters beyond molecular biology

If biological systems compute through physical organization, this idea could eventually influence several fields.

It could influence **systems biology**, by encouraging us to quantify information flow rather than only molecular abundance.

It could influence **computational biology**, by inspiring new models of cellular dynamics.

It could influence **artificial intelligence**, by suggesting architectures in which computation and physical adaptation are inseparable.

It could influence **quantum computing**, by providing biological examples of complex quantum dynamics.

And it could influence **synthetic biology**, by raising the possibility of engineering cells not merely as factories but as adaptive information-processing systems.

But these are future possibilities.

For now, the immediate scientific task is simpler:

Can we identify and measure the computational properties of biological molecular systems?

A possible research programme

If we take the hypothesis seriously, we should be able to formulate a research programme.

First, identify biological molecular systems in which information processing is clearly associated with function.

Then characterize their possible physical states.

Next, quantify the transitions among those states.

Then examine how molecular interactions modify the state space.

Then construct classical computational models.

Then, where appropriate, construct quantum models.

Then compare the predictions.

Finally, design experiments that distinguish the competing descriptions.

This approach has an important advantage.

It does not require us to assume that the quantum hypothesis is correct.

We begin with biology.

We measure the system.

We construct competing models.

We allow the data to decide.

That is a much more defensible path from speculative idea to science.

What if molecular computation is real but not quantum?

This is an important possibility.

Perhaps biological molecular computation exists, but it is entirely classical.

That would still be fascinating.

It would mean that life has evolved extraordinarily sophisticated physical information-processing architectures without needing quantum computational mechanisms.

The resulting principles could still inspire artificial systems.

Therefore, the broader hypothesis does not stand or fall with the quantum component.

There are really two questions:

Question 1: Can molecular interactions perform biologically meaningful information processing?

Question 2: Do some of those processes exploit specifically quantum information-processing effects?

The first question is broader.

The second is deeper and more speculative.

We should not confuse them.

Where my own hypothesis begins

This is the point in the series where I want to make my own intellectual position explicit.

My earlier work led me to consider the possibility that biological molecules may have richer computational properties than are usually recognized.

I proposed that molecular interactions could potentially be understood as information-processing events and, more ambitiously, that quantum computational principles might contribute to those interactions.

I also explored theoretical computational capacities of proteins based on physical limits to information processing.

These ideas are not established biological principles.

Some are considerably ahead of current experimental evidence.

I therefore do not want to present them as though the scientific community has already accepted them.

Instead, I want to ask:

Could there be a molecular layer of biological computation that is not fully captured by our current biochemical descriptions?

And, more specifically:

Could quantum physical properties contribute to that computation in some biological contexts?

These are the questions I would like to explore in the remainder of this series.

From "What if?" to "How could we know?"

This is perhaps the most important change in my own thinking.

A speculative idea is easy to formulate.

The difficult part is deciding how it could be wrong.

If we say:

"Molecules are quantum computers."

we have made a dramatic statement.

But what observation would disprove it?

If there is no conceivable observation that could disprove it, the statement is not yet a useful scientific hypothesis.

A better formulation is:

"We hypothesize that a particular molecular process may exploit quantum information processing, and therefore predict that under specified conditions it should display a measurable effect that differs from classical models."

Now we have something to investigate.

This is the standard I would like to apply to my own ideas.

The deeper question

At this stage, perhaps we can formulate the central question of the entire series more precisely.

It is not simply:

What is life?

Nor is it:

Is life quantum?

Nor even:

Are cells computers?

The question is:

How does physical matter become an information-processing, adaptive system that we recognize as living?

Chemistry provides the molecular interactions.

Organization connects those interactions.

Information describes states and relationships.

Computation describes transformations.

Feedback creates adaptation.

Networks create collective behaviour.

Evolution shapes the architecture.

And perhaps, in some cases, quantum physics contributes additional computational possibilities.

This is not yet a theory of life.

It is a way of organizing the questions.

From chemistry to information processing

The title of this article now has a more precise meaning.

We are not suggesting that chemistry should be replaced by information theory.

We are suggesting that **chemistry can be viewed at another level.**

A chemical reaction is a physical event.

A sequence of reactions can transmit information.

A network of reactions can integrate information.

A molecular state can store information.

An interaction can transform information.

A network can respond to information.

A cell can adapt to information.

At some point, the language of information processing becomes indispensable for describing what the chemistry is accomplishing collectively.

Perhaps that is the transition from chemistry to biology.

Not a replacement.

An emergence.

And perhaps this is where biology can teach computing

We often think of computer science as something that biology can borrow from.

Algorithms can help us analyse biological data.

AI can help us predict protein structures.

Machine learning can identify patterns in genomes.

Computational models can simulate biological networks.

All of this is true.

But perhaps the relationship can run in the opposite direction.

Perhaps biology has already discovered computational principles that computer science has not yet fully understood.

Biological systems compute without separating hardware from software.

They compute while changing themselves.

They compute under noisy conditions.

They compute with limited energy.

They compute while continuously interacting with their environment.

They compute through networks rather than a central processor.

And they have been doing so for billions of years.

Perhaps the future of computing will therefore involve not simply making machines faster.

It may involve learning how **living matter itself computes**.

The next step

The next article will take this question in a different direction.

If biological systems process information, and if molecular and cellular organization can generate computational behaviour, how does that compare with the systems we are building today?

Artificial neural networks.

Deep learning.

Transformers.

Large language models.

Evolutionary algorithms.

Developmental systems.

Biological learning.

There are obvious similarities.

But there are also profound differences.

Perhaps the most interesting question is not whether biology resembles AI.

It is:

Did we invent artificial intelligence—or did we rediscover, in a very limited form, principles that biology has been using for billions of years?

That is where our journey goes next.

Rethinking Biology: A Journey from Molecules to Information

Article 1 — Why Biology Needs a New Information Theory

Is biology fundamentally about molecules—or is it also a science of information?

Article 2 — Is the Cell Really Just a Bag of Chemicals?

How does organization, communication and molecular architecture transform the cell into something more than a collection of reactions?

Article 3 — A Journey from Molecules to Information

Can cellular processes be understood as information processing and computation?

Article 4 — The Quiet Revolution Called Quantum Biology

Where do genuinely quantum phenomena appear in living systems?

Article 5 — From Chemistry to Information Processing

Could molecular interactions themselves participate in biological information processing?

Coming next

Article 6 — Does Biology Compute Like Artificial Intelligence?

A final thought

A protein does not know that it is processing information.

A cell does not know that it is computing.

Evolution does not know that it is searching.

And yet information processing, computation and search may emerge from the physical organization of living matter.

That possibility changes the way we look at a molecule.

We can see it as a chemical object.

We can see it as a structural object.

We can see it as a carrier of information.

We can see it as part of a network.

And perhaps, under certain conditions, we can see it as part of a computational system.

None of these descriptions necessarily cancels the others.

They may simply describe different levels of the same reality.

The question that remains is whether these levels can be connected into a coherent theory.

Perhaps the path will look something like this:

Physics

↓

Chemistry

↓

Molecular states

↓

Interactions

↓

Information

↓

Computation

↓

Adaptation

↓

Intelligence

We are still very far from knowing whether this is the correct hierarchy.

But it gives us something valuable:

a question that can be pursued.

And perhaps the most interesting possibility is not that life violates the laws of physics.

It is that **life has learned, through evolution, to exploit the possibilities that physics provides.**

The challenge for biology is to discover how.

A note on evidence, hypothesis and my own earlier work

This article deliberately marks a transition in the *Rethinking Biology* series.

The first four articles primarily built a foundation from established biological and physical concepts. The original series plan deliberately places this article at the point where my own hypotheses begin to enter the discussion, while recommending that they be presented as questions rather than established facts.

My earlier work explored several related ideas. One study considered whether physical properties of proteins could be used to estimate theoretical limits on their computational capacity. Another explored quantum computational logic in biological networks and the possibility of both classical and quantum information processing in biological systems.

Most relevant to this article, my earlier work proposed that interacting biomolecules might be conceptualized as systems containing multiple quantum states and that their interactions could be viewed as a form of search for biologically relevant interactions.

These proposals should not be confused with established mechanisms of molecular biology.

In particular, the present article does not claim that cells have been shown to perform quantum computation, that biomolecules have been experimentally demonstrated to function as quantum databases, or that quantum search has been demonstrated as a general mechanism of molecular recognition.

Instead, I am asking whether these ideas can be reformulated into **specific, testable scientific hypotheses**.

That distinction will become increasingly important as the series moves from established science toward scientific extrapolation and, eventually, theoretical frameworks.

For me, that is the real purpose of *Rethinking Biology*:

not to replace established biology with speculation, but to ask whether today's unanswered questions can lead us toward tomorrow's testable science.